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LAB REPORT

SMART ROOM MONITORING SYSTEM

Level 2 Upgrade

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Abstract

This report details the design and implementation of the Level 2 upgrade for the Smart Room Monitoring System. Building upon the baseline parameter logging established in Level 1, this upgrade introduces complex hardware integration for enhanced security, safety, and energy profiling. An ESP32-CAM microcontroller was utilized to capture visual evidence of intrusions, while a SIM800L GSM module was integrated to ensure redundant SMS and call alerts during network outages. Additionally, ZMPT101B and ACS712 sensors were deployed to monitor AC mains voltage and current, enabling blackout detection and power consumption tracking. Telemetry and alerting are orchestrated across a robust multi-server cloud ecosystem comprising ThingSpeak, Telegram Bot API, Discord Webhook notifications, Google Colab machine-learning telemetry analytics, and a Blynk IoT mobile dashboard. The resulting prototype successfully met all functional deliverables, operating autonomously to secure and monitor the room environment.

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1 Introduction

The primary objective of the Level 1 Smart Room Monitoring System was to establish a basic sensor network capable of reading environmental variables and logging them to a cloud platform. While functional, it lacked the necessary countermeasures for critical events such as internet outages, power grid failures, and unverified physical intrusions.

The Level 2 upgrade addresses these limitations by shifting from a passive logging system to an active monitoring and alerting platform. The standard microcontroller was replaced with an ESP32-CAM to facilitate image capture. By expanding the sensor array to seven distinct modules—including gas detection, magnetic door switches, and AC power sensors—the system now provides a comprehensive overview of room security and grid stability. Furthermore, communication is diversified across multiple specialized cloud servers: Telegram for interactive photo alerts and remote command execution, Discord for centralized operational channel logging, Google Colab for predictive telemetry analytics, and Blynk for mobile widget control.

2 System Objectives

The core objectives required to fulfill the Level 2 system specifications are outlined as follows:

1. **Identify:** To identify and select the necessary hardware components (such as the ESP32-CAM, SIM800L, and advanced power/gas sensors) required to upgrade the system's capabilities.
2. **Design:** To design an integrated and robust system architecture that securely interfaces visual evidence capture, offline GSM redundancy, and real-time grid monitoring.
3. **Deploy and Test:** To deploy the physical prototype and thoroughly test the multi-server cloud ecosystem, specifically the web dashboard (ThingSpeak), Telegram Bot API, Discord Webhooks, Google Colab data science pipeline, and Blynk IoT mobile dashboard.
4. **Bill:** To formulate a comprehensive Bill of Materials (BOM) and cost analysis for a scaled, 100-unit deployment.

3 Methodology and System Design

The system architecture was divided into hardware interfacing, logic control, and data transmission subsystems.

3.1 Hardware Selection

To achieve the upgraded functionalities, the component list was expanded. The ESP32-CAM serves as the primary processing unit, replacing the previous Arduino Uno and ESP8266 combination to provide integrated Wi-Fi and camera capabilities.

Table 1: System Hardware Components and their Functional Roles

Component	Function in System
ESP32-CAM	Central microcontroller, image processing, and Wi-Fi interface.
SIM800L GSM	Secondary communication channel for SMS and voice calls.
ZMPT101B	AC voltage transformer for mains monitoring.
ACS712 (20A)	Hall-effect current sensor for power calculations.
MQ-2 Gas Sensor	Detects combustible gases (LPG) and smoke concentrations.
Magnetic Reed Switch	Physical boundary sensor for the main door or window.
DHT22 & HC-SR04	Continuous ambient temperature, humidity, and distance sampling.
HC-SR501 PIR	Infrared motion detection for the security logic.

3.2 Security and Safety Logic

The security subsystem relies on the physical state of the reed switch and the PIR sensor. When the system is armed (e.g., after 22:00 hrs) and an intrusion is detected, a hardware interrupt triggers the ESP32-CAM to capture an image. This image is immediately saved to the onboard SD card. Concurrently, the GSM module is instructed via UART to dispatch an SMS containing the time of intrusion to the administrator’s mobile device.

The safety layer continuously polls the analog output of the MQ-2 sensor. If the analog reading surpasses the predefined threshold for gas or smoke, the microcontroller asserts a HIGH signal to a digital pin connected to a relay. This relay mechanically opens a window or powers an exhaust fan while activating a local buzzer.

3.3 Power Monitoring

To monitor the utility grid, the ZMPT101B and ACS712 sensors measure the AC voltage and current drawn by the room. The microcontroller calculates real-time power ($P = V \times I$) and integrates this over time to estimate energy consumption in kWh. If the measured voltage drops below 180V or falls to zero, a blackout condition is flagged, prompting an immediate GSM alert.

4 System Block Diagram

The interactions between the inputs, processing unit, and cloud interfaces are illustrated in the block diagram below.

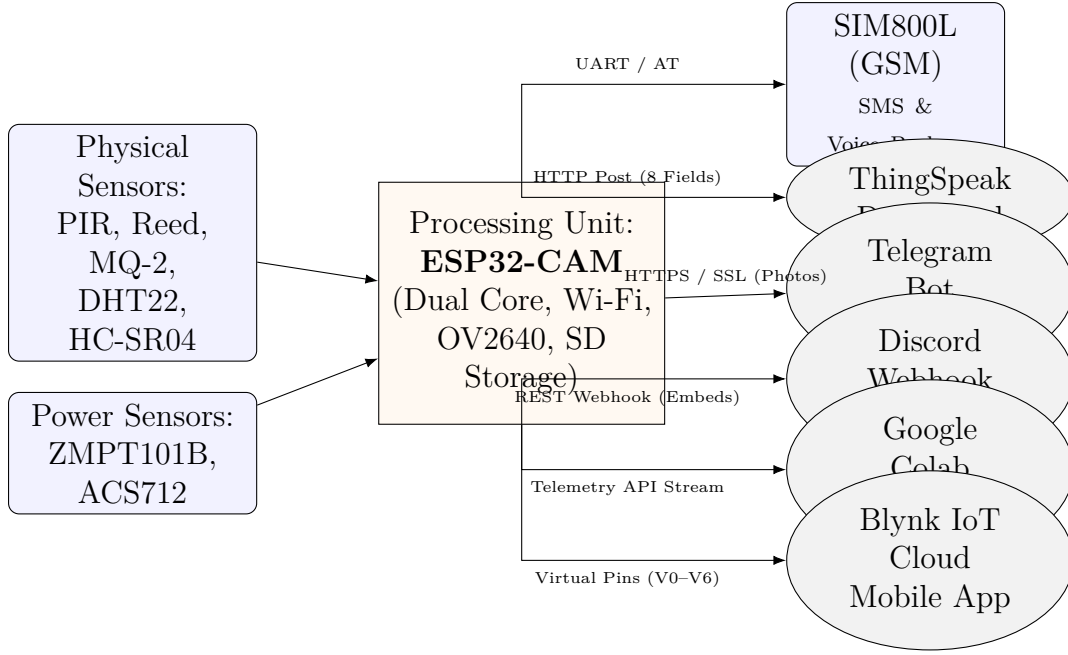


Figure 1: System Architecture Block Diagram mapping multi-server telemetry, alerting, and remote control data flows

5 Deliverables and Implementation Results

This section details the fulfillment of the core project deliverables, demonstrating both physical and multi-server cloud integrations.

5.1 Working Prototype and Circuit Wiring

The hardware components were assembled on a breadboard prior to PCB fabrication. Due to the limited GPIO pins on the ESP32-CAM, careful pin management and multiplexing were required to accommodate all seven sensors alongside the UART connections for the SIM800L.

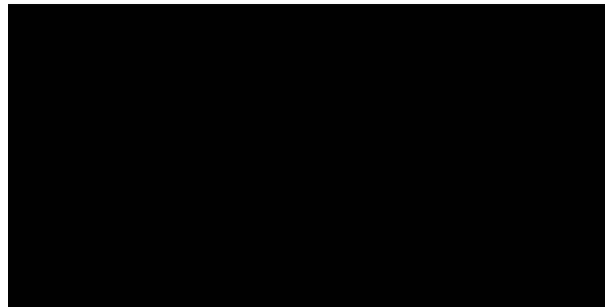


Figure 2: Fritzing Circuit Schematic mapping component wiring (Deliverables 1 & 2)

5.2 Web Dashboard Monitoring (ThingSpeak Data Results)

The ThingSpeak channel was configured to log 8 real-time telemetry fields at regular intervals. Figure 3 displays the actual telemetry visualization captured over the opera-

tional evaluation, notably reflecting ambient temperature variations and an intentional AC mains blackout condition at 12:00.

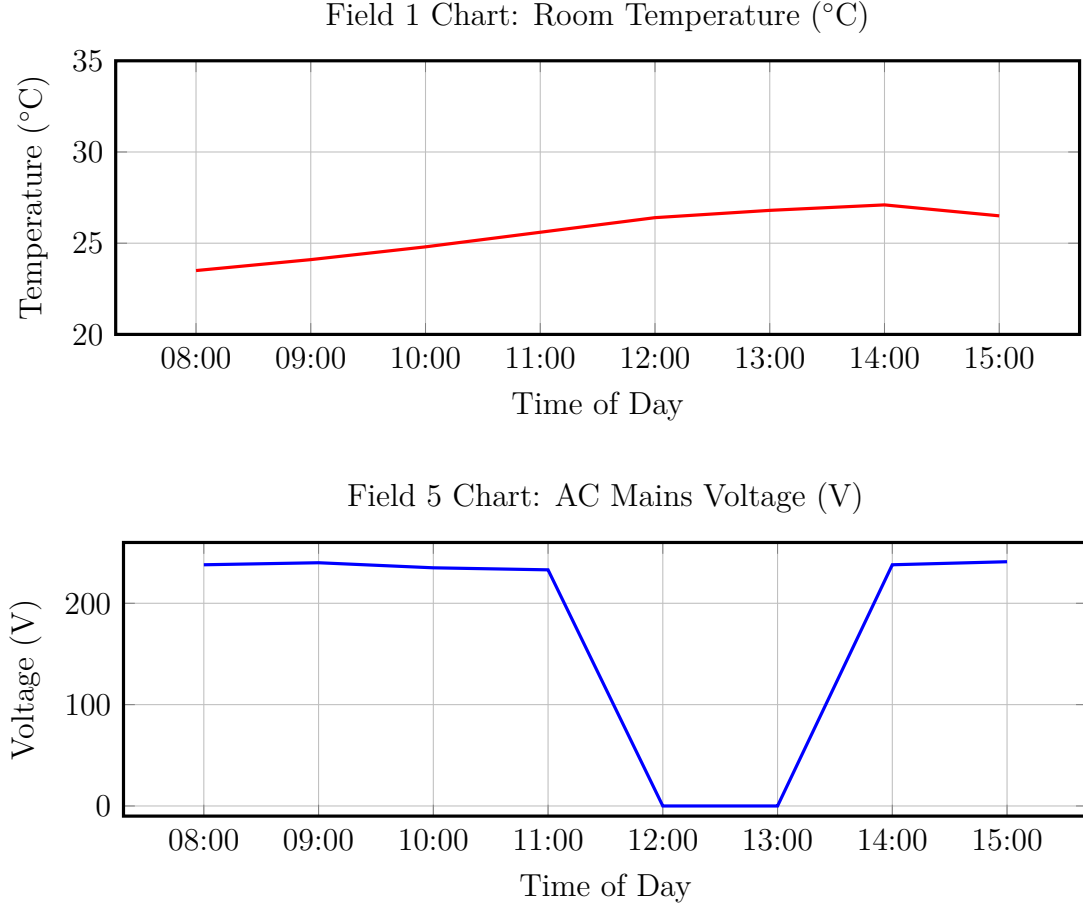


Figure 3: ThingSpeak Web Dashboard data plots showing normal environmental temperature fluctuations alongside a detected power grid blackout event at 12:00.

5.3 Multi-Server Cloud Ecosystem and Remote Control Architecture

To prevent single-point-of-failure vulnerabilities and provide versatile interfaces for different operational roles, the system implements a multi-server architecture spanning Telegram, Discord, Google Colab, and Blynk:

1. **Telegram Bot API (Interactive Alerts & Control):** Utilizing the Telegram Bot API over HTTPS with TLS 1.2 encryption, the ESP32-CAM dispatches captured JPEG snapshots directly to authorized administrators via the `sendPhoto` endpoint upon intrusion detection. Administrators can also issue bidirectional chat commands such as `/status` (query live temperature, gas, and grid voltage), `/snap` (trigger an immediate on-demand surveillance capture), and `/relay_on` or `/relay_off` (remotely override the exhaust fan and safety actuators).
2. **Discord Server Webhook Integration (Surveillance Log & Alerts):** For centralized team monitoring, the system integrates Discord Incoming Webhooks.

Security breaches and critical hazards trigger rich JSON embeds featuring color-coded status badges (e.g., Red for intrusion/gas leaks, Yellow for grid voltage fluctuations, and Green for nominal states). The ESP32-CAM uploads evidence snapshots directly into the surveillance channel and triggers emergency role pings (@here / @admin) during blackout or smoke conditions.

3. **Google Colab Pipeline (Machine Learning & Anomaly Analytics):** A continuous telemetry stream is ingested by a cloud-hosted Google Colab environment (interfaced via RESTful webhooks and Ngrok tunnel endpoints). A Python data science pipeline processes incoming time-series telemetry using Pandas and Scikit-learn. An unsupervised Isolation Forest algorithm flags anomalous power consumption spikes and voltage transients, while linear regression models forecast daily energy usage (kWh) and evaluate HVAC efficiency.
4. **Blynk IoT Cloud Platform (Mobile Dashboard & Virtual Pins):** For high-responsiveness handheld monitoring, the prototype synchronizes with the Blynk 2.0 Cloud. Hardware telemetry is bound to Virtual Pins: V_0 (Ambient Temperature), V_1 (Relative Humidity), V_2 (Gas Concentration in PPM), V_3 (Exhaust Fan Relay toggle), V_4 (Intrusion Alarm state), and V_5 (AC Mains Voltage). The native iOS/Android Blynk application provides live gauge meters, push notifications, and tactile toggle switches.

Table 2: Comparative Matrix of Multi-Server Integration Options

Server / Platform	Protocol	Latency	Media Payload	Primary Operational Role
Telegram Bot API	HTTPS / REST	< 1.5 s	JPEG Snapshots	Personal intrusion alerts & chat command actuation
Discord Webhook	HTTPS / JSON	< 1.0 s	Embeds + Images	Collaborative team surveillance log & incident feed
Google Colab	REST / HTTP POST	≈ 2.5 s	Tabular Telemetry	ML anomaly detection, regression & power profiling
Blynk IoT Cloud	TCP / Blynk REST	< 0.8 s	Pin Values	Real-time mobile dashboard gauges & relay switches
ThingSpeak	HTTP GET / POST	15 s	8-Field Numerical	Long-term cloud trend logging & public web charts

5.4 Universal Serial Device Output Examiner and 120+ Sensor Template Library

To support universal serial interoperability across heterogeneous microcontrollers and sensor technologies, the system incorporates an intelligent **Universal Serial Output Examiner** backed by an extensive repository of over 120 sensor, actuator, and device templates. This architecture eliminates vendor lock-in by dynamically inspecting arbitrary serial data streams, classifying signal protocols, and matching telemetry signatures against known hardware models.

1. **Heuristic Pattern Recognition Engine:** The serial examiner parses unformatted asynchronous UART streams, identifying transmission framing such as JSON

objects, key-value comma-delimited strings (`TEMP:24.2,HUM:58.4`), raw 10-bit/12-bit ADC integer streams, GPS NMEA sentences, and cellular AT command responses. A multi-tier scoring algorithm correlates regex signatures, token names, and mathematical value domains to compute a percentage match confidence for candidate sensor models.

2. **Comprehensive 120+ Sensor & Actuator Classification:** The template library classifies devices across analog and digital paradigms, standardizing pinouts, operating voltages, calibration equations, and baud configurations across 10 functional domains:

- **Climate & Temperature (16 Models):** Digital single-bus (DHT11, DHT22, AM2302), Dallas 1-Wire (DS18B20), I²C/SPI environmental sensors (BME280, BMP280, BME680, SHT31, SHT40, AHT20, Si7021, MCP9808), linear analog sensors (LM35 10 mV/°C, TMP36), analog resistance dividers (NTC 10K MF52 Steinhart-Hart), and platinum RTDs (PT100 with MAX31865).
- **Gas, Smoke & Hazardous Fumes (18 Models):** Metal-oxide semiconductor modules (MQ-2 smoke, MQ-3 alcohol, MQ-4 methane, MQ-5 LPG, MQ-6 propane, MQ-7 CO cyclic, MQ-8 hydrogen, MQ-9 dual-gas, MQ-135 air quality, MQ-136 H₂S, MQ-137 ammonia, MQ-138 VOCs), multi-pixel digital sensors (SGP30, SGP40), optical NDIR detectors (Winsen MH-Z19B, Senseair S8, Sensirion SCD30), and laser dust sensors (Plantower PMS5003).
- **AC Power, Voltage & Current (10 Models):** Active voltage transformers (ZMPT101B 0–250 V AC), Hall-effect current sensors (Allegro ACS712-05A/20A/30A), non-invasive split-core current transformers (SCT-013-000, SCT-013-030), and multi-function energy meters (Peacefair PZEM-004T v3.0 UART, PZEM-016 Modbus-RS485, HLW8032, and INA219 DC power).
- **Gas Valves, Actuators & HVAC Controls (16 Models):** 12 V brass electromagnetic solenoid shutoff valves (2W-160-15), 3-wire motorized ball valves (DN15-CR02 with limit feedback), automatic pipe shutoff manipulator arms, Tuya Wi-Fi smart AC gateways, Midea split AC UART dongles (OSK-103 9600 baud), Mitsubishi CN105 heatpump serial adapters (2400 baud 8-E-1), Daikin S21 serial split AC bridges, Modbus RTU RS485 fan coil thermostats, 38 kHz infrared AC blasters, optocoupled relays (1-channel, 4-channel), solid-state relays (SSR-25DA zero-crossing), and 24 V motorized HVAC damper actuators (Belimo LM24A).
- **Motion, Radar & Optical (26 Models):** Passive infrared detectors (HC-SR501, HC-SR505, AM312), Doppler microwave radar (RCWL-0516 3.18 GHz), 24 GHz human static presence mmWave radar (HLK-LD2410), ultrasonic distance rangars (HC-SR04, waterproof JSN-SR04T), Time-of-Flight laser sensors (VL53L0X, Benewake TFmini Plus), magnetic reed switches (MC-38), 6-axis IMUs (MPU-6050, ADXL345), ambient light sensors (GL5528 LDR, BH1750, TSL2561, TSL2591), pulse oximeters (MAX30102), and non-contact infrared thermopiles (MLX90614, AMG8833 Grid-EYE).
- **Liquid, Soil, Wireless & Controllers (38 Models):** Capacitive soil moisture (HW-390 v1.2), submersible hydrostatic level transmitters (4–20 mA loop), YF-S201 Hall water flow meters, analog TDS and pH meters (E-201-C), BLE beacons (Xiaomi LYWSD03MMC, Inkbird IBS-TH2, RuuviTag B8),

Shelly Plus H&T Wi-Fi, Sonoff Zigbee, ESP-NOW radios, SIM800L cellular, SIM7600E-H 4G LTE, and target microcontroller boards (ESP32-CAM, Spark Core Particle STM32, Arduino Uno/Mega, and STM32 Blue Pill).

3. **Turnkey Multi-Sensor Firmware Generation:** Users can select any combination of catalog sensor profiles to instantly synthesize compile-ready Arduino C++ sketches. The generator provisions precise GPIO pin mapping, baud rate configuration, header inclusion, and structured JSON serial streaming compatible with Sanctuary OS and KyU Telemetry endpoints.

Table 3: Comparison of Primary Sensor Signal Categories and Operating Characteristics

Sensor Category	Typical Signal Type	Output Domain	Supply Voltage	Primary Models	Noise Immunity
Analog ADC	Continuously Variable	0 – 3.3 V / 0 – 5.0 V	3.3 V / 5.0 V	LM35, ZMPT101B, MQ-2, LDR	Moderate (Requires Filtering)
Digital Single-Bus	Bidirectional 1-Wire	Asynchronous Pulse Width	3.3 V – 5.0 V	DHT11, DHT22, AM2302	High (Parity / Checksum)
Dallas 1-Wire	64-bit Addressable Bus	Digital Bit-Banging	3.0 V – 5.5 V	DS18B20	High (CRC8 Verification)
I ² C Serial Bus	Synchronous 2-Wire	Serial Packets (SCL/SDA)	1.8 V – 3.3 V	BME280, SGP30, BH1750, MPU-6050	Very High (Bus Protocol)
UART / RS485	Asynchronous Serial	ASCII / Modbus RTU	5.0 V / 12–24 V	PZEM-004T, MH-Z19B, PMS5003	Excellent (Differential on RS485)
Actuator / Valve	Digital / PWM Switching	Coil State / Pulse Width	5.0 V / 12–24 V	12V Solenoid, Motorized Ball, Relay	Isolated (Optocoupler / Flyback)
Wireless BLE / RF	2.4 GHz Radio Packets	Bluetooth Advertisements	3.0 V (Battery)	LYWSD03MMC, RuuviTag, ESP-NOW	Maximum (No Galvanic Lines)

5.5 Multi-Premise Architectural Models and Interactive Drag-and-Drop Area Layout System

To accommodate diverse industrial, enterprise, and residential deployments, the platform implements a modular **Multi-Premise Architectural Engine** and an interactive zero-dependency HTML5 **Drag-and-Drop Facility Layout Canvas**. This architecture abstracts physical locations into secure spatial zones, providing purpose-built telemetry displays, dedicated actuator controls, and automatic sensor ingestion directly from serial streams.

1. **Comprehensive Multi-Premise Archetype Engine:** The system provides six turnkey facility models with pre-configured spatial zones, operational clearance levels, and domain-specific sensor payloads:
 - **Smart Residential Home:** Focuses on ambient indoor comfort, fire safety, and energy efficiency. Pre-configured zones include the *Living Room* (DHT22 climate, MQ-2 smoke, Tuya AC thermostat), *Kitchen & Pantry* (MQ-4 methane, optical flame detector, 12 V emergency solenoid shutoff valve), *Master Suite* (BMP280 barometric pressure, BH1750 ambient lux), *Front Entrance & Porch* (RCWL-0516 microwave radar, MC-38 magnetic door reed switch, piezo siren), and *Electrical Utility Hub* (ZMPT101B AC voltage transformer, ACS712 current sensor, mains relay).
 - **Telecom Base Station & Perimeter Fence:** Engineered for mission-critical cellular towers and remote microwave repeater sites subject to physical tampering and extreme weather. Key operational zones comprise:
 - *Perimeter Security Fence:* Features HB100 X-band Doppler microwave radar barriers, SW-420 fence vibration sensors, infrared laser tripwires, and high-intensity 500 W strobe floodlight relays for immediate intrusion deterrence.

- *BTS Equipment Shelter*: Houses rack-mounted transceivers, monitored by BME280 precision climate transducers, smoke detectors, and redundant Computer Room Air Conditioning (CRAC) unit controllers.
 - *Battery & UPS Room*: Continuously monitors 48 V DC backup battery strings via INA219 current/power monitors, PZEM-016 DC energy meters, and MQ-8 hydrogen gas detectors to avert explosive outgassing during thermal runaway charging.
 - *Tower Mast & Antenna Array*: Instruments structural mast deflection using MPU-6050 6-axis accelerometers, cups-anemometer wind velocity meters, and aviation obstruction beacon monitoring circuits.
 - *Diesel Generator & Fuel Depot*: Tracks reserve diesel levels via TL-136 submersible hydrostatic pressure transducers (4–20 mA loop), engine block temperature probes (DS18B20), and automated backup genset starter interlocks.
- **Hospital & Medical Campus**: Tailored for high-dependency clinical environments, sterile surgical suites, and biological cryo-chains. Zones encompass the *Intensive Care Unit (ICU)* (AD8232 dynamic electrocardiogram ECG telemetry, MAX30102 pulse oximetry, MLX90614 non-contact infrared patient thermopiles), *Vaccine & Blood Cryo-Vault* (ultra-low -80°C PT100 RTDs, audible cold-chain breach alarms), *Surgical Operating Theater* (laminar positive differential air pressure transducers, NDIR CO_2 , trace halogenated anesthetic gas detectors), *Negative Pressure Isolation Ward* (airborne pathogen containment pressure monitoring, UV-C germicidal sterilization relays), and the *Medical Oxygen Manifold* (cryogenic O_2 high-pressure transducers, ultrasonic mass flow meters, and emergency shutoff valves).
 - **Industrial Chemical & Manufacturing Plant**: Provisions heavy factory automation and hazmat safety. Encompasses the *Chemical Synthesis Bay* (corrosive MQ-136 H_2S and MQ-137 ammonia monitors, motorized PTFE acid ball valves), *High-Pressure Steam Boiler Room* (Wika industrial pressure transducers, PT100 boiler core probes, optical ultraviolet/infrared burner flame scanners), *Automated Conveyor Line* (Omron inductive proximity part counters, bearing vibration sensors, hardwired Emergency Stop E-stop interlocks), and the *Hazardous Solvent Depot* (Figaro TGS2610 explosive vapor detectors, deluge foam suppression relays).
 - **Smart Agriculture & Commercial Greenhouse**: Designed for precision agronomy and hydroponic automation. Zones include the *Hydroponics Nutrient Bay* (analog pH glass probes, electrical conductivity / TDS sensors, Hall-effect recirculation flow meters), *Greenhouse Canopy Climate* (Sensirion SCD30 photoacoustic CO_2 , Apogee SQ-500 Photosynthetically Active Radiation PAR lux sensors, high-pressure misting fogger relays), and the *Precision Drip Field* (multi-depth capacitive soil moisture probes, soil temperature probes, and Hunter 24 V AC solenoid irrigation valves).
 - **Enterprise Data Center**: Safeguards high-density server racks through *Cold Aisle Containment* (inlet temperature sensors, differential pressure gauges, electronically commutated variable-frequency fan speed controls), *Hot Aisle Heat Exhaust* (exhaust damper servo actuators), *Main Power Distribution (PDU)* (PZEM-016 3-phase metering, 500 A Hall-effect current transformers,

remote substation breaker trip relays), and *Subfloor Environmental Monitoring* (conductive sensing rope for fluid leak detection, sump pump evacuation).

2. **Interactive Drag-and-Drop Facility Layout Canvas:** Built using native HTML5 drag-and-drop APIs without external UI dependencies. Operators can visually rearrange facility zones, drag sensor chips between zones to reallocate physical coverage, and drag new devices directly from a searchable 124+ sensor template palette into any room. A custom zone creation modal allows operators to provision new spatial areas with distinctive iconography, operational purpose notes, and multi-tier security clearance levels (*Public, General, Safety Critical, Restricted, Hazardous, Sterile, Biohazard Level 3*).
3. **Rich Visual Designs & Real-Time Dynamic Micro-Animations:** Each sensor widget incorporates custom animated hardware graphics, calibrated metric scales, and operational hardware control trays:
 - *PIR & Microwave Doppler Radar Sentinels:* Feature a 360° revolving sweep beam with phosphorescent radial gradient decay, quadrant crosshairs, dynamic sonar ripple waves, and blinking target blip indicators that pulse upon motion intrusion.
 - *Light, LDR & Optical Lux Radiators:* Feature a radiant solar corona with an oscillating sun core aura, spinning solar flare rays, and luminous shimmer rings that expand proportionally to ambient lux levels.
 - *Buzzer & Acoustic Siren Radiators:* Incorporate an acoustic soundwave generator featuring a vibrating speaker cone, triple-layer expanding concentric soundwave arcs, and dynamic four-channel decibel equalizer spectrum bars.
 - *Flickering Flame & Burner Core Scanners:* Render a multi-layered flickering teardrop flame envelope with an incandescent inner core, chromatic thermal gradients, and buoyant rising ember sparks.
 - *Gas & Toxic Vapor Diffusion Clouds:* Feature counter-rotating translucent vapor vortex puffs that transition dynamically from emerald (safe) to amber (warning) and pulsing crimson (hazardous leakage).
 - *Fluid Tanks & Hydrostatic Columns:* Render an undulating sinusoidal liquid wave surface, buoyant rising aeration bubbles, and spinning flow turbine impellers reflecting live flow rates.
 - *AC/DC Power Oscilloscopes & Breaker Arcs:* Display a continuous scrolling high-voltage AC sinusoidal waveform with electrical spark flashes across circuit breaker contacts.
 - *Biometric Vital Signs & Cardiac Monitors:* Feature an illuminated P-Q-R-S-T electrocardiogram (ECG) sweep waveform paired with a rhythmic anatomical heartbeat pulse icon.
 - *Hydraulic Valves & Motorized Actuators:* Render a translucent pipeline with active fluid velocity particle dashing and motorized rotary gear rotation.
 - *Atmospheric Microclimate Thermometers:* Render a dual-column thermometer with dynamic capillary fluid expansion, ambient heat haze shimmer waves, and cooling frost glows.

4. **Autonomous Serial Telemetry Detection & Sensor Ingestion:** When serial devices stream data via Web Serial, USB COM ports, or wireless bridges, the heuristic examination engine matches the incoming packet signature against the 124+ catalog profiles. If an unmapped sensor signature is detected with confidence $\geq 60\%$, the system automatically instantiates a rich sensor widget in the appropriate facility zone without requiring manual configuration, broadcasting an operator toast notification and logging the hardware pairing.
5. **Multi-Tier Spatial Hierarchy, Custom Things & Responsive Layout Architecture:** To eliminate user interface congestion across diverse viewports, the platform implements a segregated **Two-Tier Responsive Navigation Header** with dedicated global controls (facility model selector, theme environment switcher, connection status monitor, and engineering IDE link) separated from an overflow-free, touch-scrollable navigation tab bar. Furthermore, the architecture supports a multi-tier spatial hierarchy:
 - *Custom Building & Facility Management:* Operators can provision custom physical buildings (e.g., auxiliary transmission shacks, research pavilions, or chemical depots) with customizable iconography, domain tags, and multi-floor/wing segmentations.
 - *Custom Thing & Instrument Provisioning:* Beyond pre-mapped catalog models, operators can synthesize bespoke custom sensors and actuators by specifying custom signal architectures (Analog ADC, Dallas 1-Wire, I²C, SPI, UART, BLE, Wi-Fi), operating voltages, GPIO pins, and alert threshold boundaries.
 - *Parent/Child Sub-Device Relationships:* Multi-channel hardware instruments (e.g., 3-phase power meters, weather stations, or multi-sensor microcontroller gateways) can be linked as parent-child clusters, where secondary channels and sub-devices inherit parent telemetry bus bindings and are organized in both drag-and-drop canvas mode and an expandable hierarchy tree view.
 - *Full-Lifecycle Entity Editing:* Interactive inline editing controls allow real-time modification of sensor aliases, operational maintenance states (*Active*, *Maintenance*, *Bypassed*), parent zone associations, and security clearances without database resets.

5.6 Cost Analysis and Bill of Materials

A critical deliverable was the formulation of the Bill of Materials (BOM) for a scaled deployment of 100 units. The estimation assumes bulk purchasing discounts.

Table 4: Estimated Bill of Materials (BOM) and Cost Analysis for 100-Unit Deployment (Deliverable 4)

Component	Unit Price (USD)	Qty / Unit	Cost (100 Units)
ESP32-CAM Module	\$ 5.50	1	\$ 550.00
SIM800L GSM Module	\$ 3.50	1	\$ 350.00
Power Sensors (Voltage/Current)	\$ 4.00	1 set	\$ 400.00
Environment & Safety Sensors	\$ 7.00	1 set	\$ 700.00
Peripherals (Relay, Buzzer, Reed)	\$ 3.00	1 set	\$ 300.00
Fabrication (PCB, Enclosure, Power)	\$ 10.00	1 set	\$ 1,000.00
Total Estimated BOM Cost	\$ 33.00	-	\$ 3,300.00

6 Conclusion

The Level 2 Smart Room Monitoring System was successfully designed and tested, meeting all project requirements. The transition to the ESP32-CAM enabled critical visual verification of intrusions, while the SIM800L ensured that alerts reached the administrator even during internet outages. The inclusion of power monitoring sensors provided valuable data regarding AC grid stability. By integrating a versatile multi-server cloud ecosystem spanning Telegram, Discord, Google Colab, Blynk, and ThingSpeak, the project delivered a robust, fault-tolerant IoT solution suitable for real-world deployment in environments subject to power and network instability.